

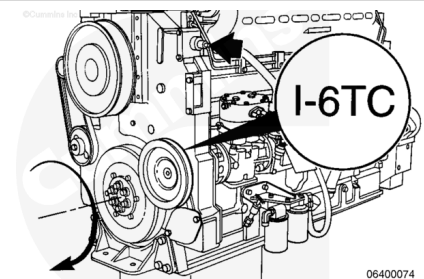
Trainings ▾

Preparatory Steps

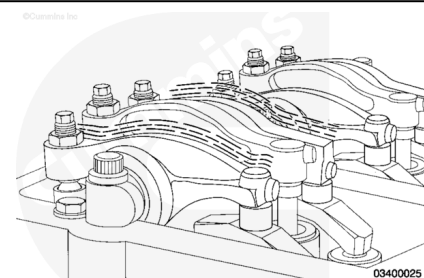
- Remove the rocker lever cover on Number 1 cylinder. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html)



- The barring device shaft turns approximately two revolutions before the engine begins to turn. The device will **not** turn the engine opposite the direction of normal rotation.
- Push the shaft in and turn the barring device counterclockwise until the "I-6TC" mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.



- If both the intake and exhaust number 1 cylinder rocker levers wiggle and the mark on the accessory pulley align with the "I-6TC" mark, then the Number 1 cylinder is on its compression stroke. If the marks are **not** aligned, rotate the crankshaft one full revolution to place the Number 1 cylinder on its compression stroke.



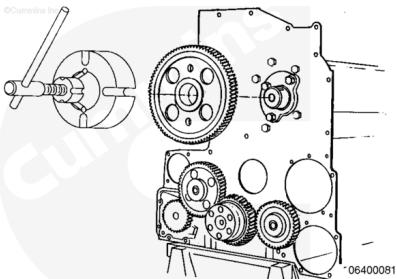
Remove

Use the camshaft gear puller kit, Part Number 3824900, to remove the camshaft gear.

Attach the puller to the gear.

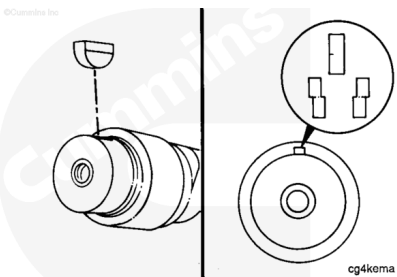
Remove the gear.

Remove the puller.



Check and record the direction of the key offset when compared to the camshaft gear rotation (same or opposite).

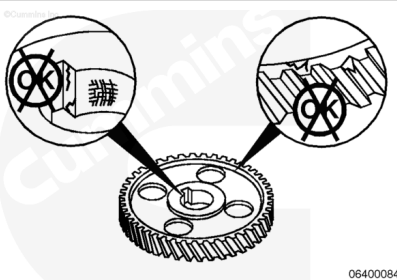
Remove the key.



Inspect for Reuse

Inspect the camshaft gear for cracks, chipped or broken teeth.

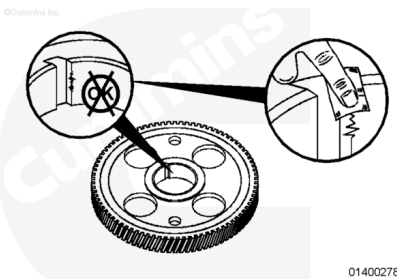
Inspect the bore of the gear for fretting or burrs.



Inspect the gear keyway for burrs.

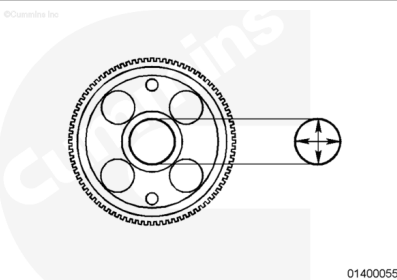
Remove any burrs with fine crocus cloth.

If the keyway is damaged or it is **not** possible to remove the burrs with fine crocus cloth, the gear **must** be replaced.



Measure the inside diameter of the camshaft gear.

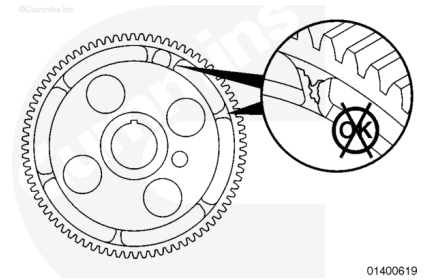
Camshaft Gear Inside Diameter		
mm		in
57.124	MIN	2.249
57.150	MAX	2.25



The gear **must** be replaced if the inside diameter is **not** within specifications.

With Position Sensor

Inspect the backside of the gear for cracks, chipped, or broken bosses that the cam position sensor reads. The gear **must** be replaced, if the bosses are damaged.

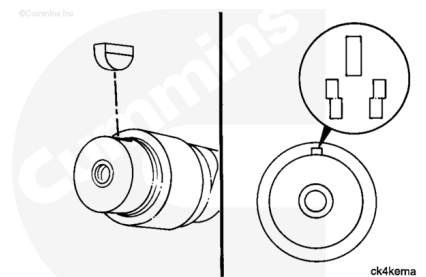


Install

Camshaft keys are available in different sizes (amount of offset). The injection timing is controlled by:

- the selection of the camshaft key
- the direction of the key offset when compared to the camshaft gear rotation
- the amount of offset.

Install the same part number key in the same orientation as the key that was removed. Refer to Procedure 006-025 in Section 6 (</qs3/pubsys2/xml/en/procedures/20/20-006-025-tr.html>) for further information on camshaft key selection.



⚠ CAUTION ⚠

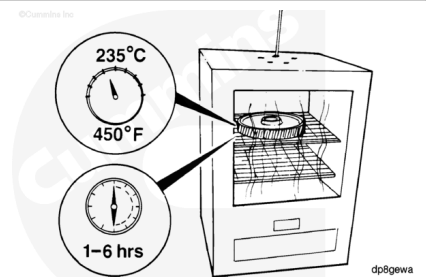
Wear protective clothing to prevent personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

Do not attempt to install the gear without using heat.



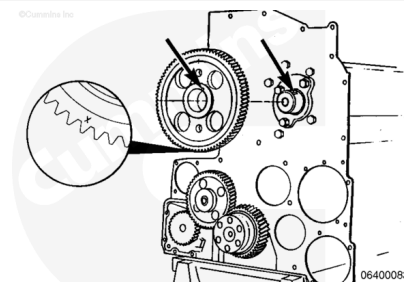
Use an oven and adjust the heat to 235°C [450°F]. Heat the gear for a **minimum** of 1 hour, but no more than a maximum of 6 hours. The inside diameter of the gear will become larger and will simplify installation.

⚠ CAUTION ⚠

Wear protective clothing to prevent personal injury from burns.

⚠ CAUTION ⚠

Allow the gear to cool slowly. Do not use water or oil to reduce the cooling time. This will cause the gear to crack.

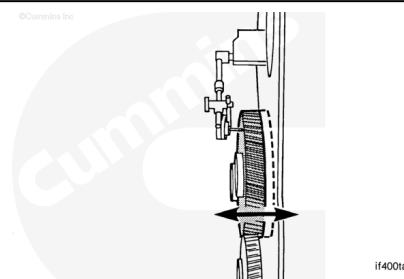


Note : The timing mark on the camshaft gear **must** be visible from the front of the gear after it is installed on the camshaft.

Remove the gear from the oven and install it on the camshaft. The keyway in the gear **must** be aligned with the key in the camshaft.

Use a dial indicator and measure the camshaft end clearance.

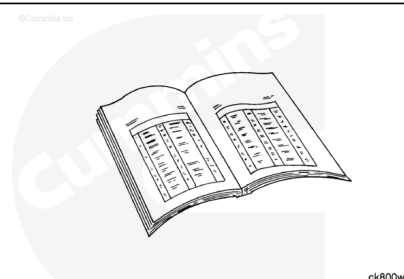
Camshaft End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013



If the clearance is too small, check for foreign material or a piece of gasket between the thrust plate and the block.

Finishing Steps

- Install the rocker lever cover of Number 1 cylinder. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1.



(/qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html)

- Install the gear cover and all related components. Refer to Procedure 001-031 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

Last Modified: 17-Sep-2015
